ACADEMIC YEAR: 2025-26

Course: Skill base Lab Course: Cloud Computing

Course code: CSL605

Year/Sem: TE/VI

Experiment No.: 05
Aim:To study and Implement Storage as a Service using AWS S3.
Name: Shravani Sandeep Raut
Roll Number: 40
Date of Performance:09/02/2026
Date of Submission: 11/02/2026

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission.	10	
Total	**20**	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	4-5	2-3	2
Understanding	4-5	2-3	2
Journal work and timely submission.	8-10	5-8	4

Checked by

Name of Faculty : Dr. Tatwadarshi Nagarhalli

Signature :

Date :

Experiment No. 5

Aim: To study and Implement Storage as a Service using OwnCloud/ AWS S3, Glaciers/ Azure Storage.

Objective: Understand the concept of SaaS and implement using Own Cloud which gives universal access to files through a web interface.

Theory:

- SaaS is defined as the software distribution model deployed on the internet in which a cloud service provider provides applications.
- It is also known as "on-demand software" or "pay-as-you-go application". Here the customer licenses their product via SaaS-providers.
- SaaS market is a rapid-growing one, and with this fast-growing service, SaaS will soon become an active cloud service technology for every organization and company.
- In SaaS, the software & the applications associated with it are centrally located on the cloud server. Users can access them via a thin client connecting application, i.e., using a web browser.
- The SaaS provides various applications such as: CRM applications, Solution to Human Resource (HR), Pre-existing Billing & Invoicing systems, Other daily usable application suites.
- <u>Advantages of SaaS</u>
    - Easy to buy: SaaS's cost is based on monthly or yearly fees allowing new organizations to access the world of business at a low-cost, at least lesser than licensed application.
    - Minimization of Hardware Requirement: All SaaS software is hosted remotely & so there is no or lesser need for hardware for the organizations.
    - Special Software: No special software versions are required, as all the users will use the same software version. SaaS reduces IT costs by outsourcing hardware & software maintenance.
    - Low Maintenance: SaaS removes the daily problem of installing, maintaining, and updating software. The set-up cost of SaaS is also less in comparison to enterprise software.
- <u>Disadvantages of SaaS</u>
    - Latency factor: comes due to a variable distance of data between the cloud & the end-user, and hence a possibility of latency may arise while interacting with applications.
    - Internet Connection: is a major issue. Without an internet connection, SaaS applications are unusable.
    - Switching between SaaS vendors in case of any change is very difficult.
    - The SaaS cloud service is not very secure as in-house deployment.

Steps:

Step-1: click on create bucket

Step-2: Give Bucket name & select region for storage

Step-3: Keep object ownership setting as ACLs Disabled as

by-default Step-4: Disable block all public access

checkbox

Step-5: Select the checkbox for Turning off block all public access might result in this bucket and the objects within becoming public

Step-6: Keep bucket versioning as disabled and add tags if required.

Step-7: Keep default encryption disabled and click on create bucket button

You can now see the successful creation of your bucket

Step-8: now click on the bucket that you have created

Step-9: You can either create a folder here or upload an existing file in the bucket

Step-10: now click on upload button and click on add files button browse your local machine and select which file you need to upload on S3 next click on upload button at bottom right end

Now you can check the upload status screen

Now click on close button

Step-11: Select properties and scroll down to Static website hosting option which is disabled now click on Edit option on right side

Step-12: Enable the radio button and specify the file name in Index document which you have added in S3

Scroll down and save the changes at bottom

right Step-13: Click on Permissions Tab

Step-14: In bucket policy click on Edit option

Step 15- after clicking on edit button paste the following code in bucket policy

```
{
        "Version": "2012-10-17",
        "Statement": [
```

```
{ "Sid": "PublicReadGetObject",
"Effect": "Allow",
 "Principal":
 "*", "Action": [
        "s3:GetObject"
         ],
         "Resource": [
                "arn:aws:s3:::Bucket-Name/*"
                ]
        }
]
}
```

Note-Make sure that you add your bucket name in the code above

Scroll down and click on Save Changes button

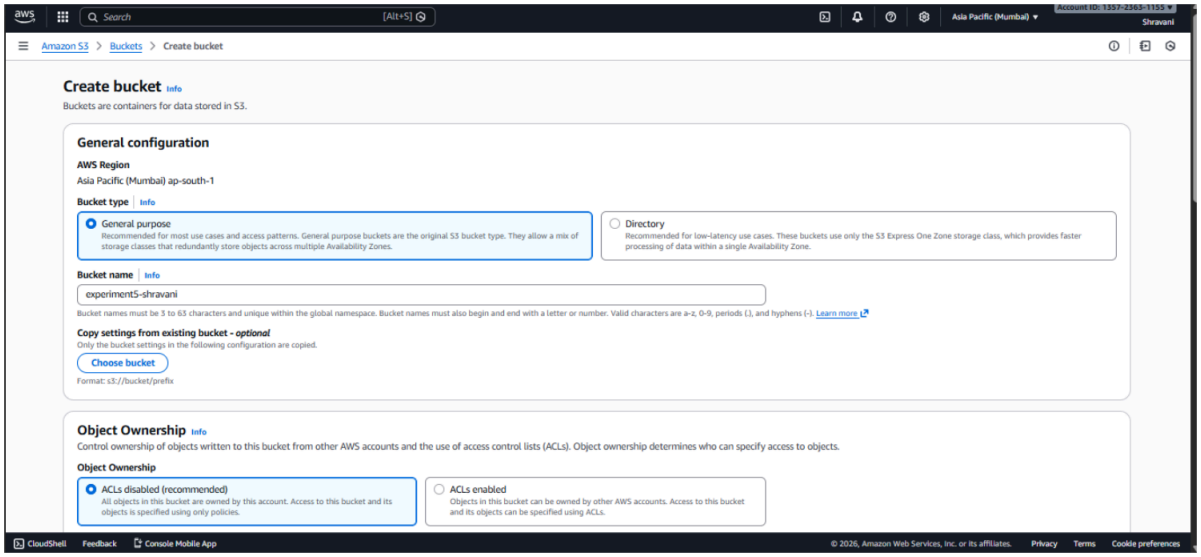
Step-16: open your html file and click on Object URL

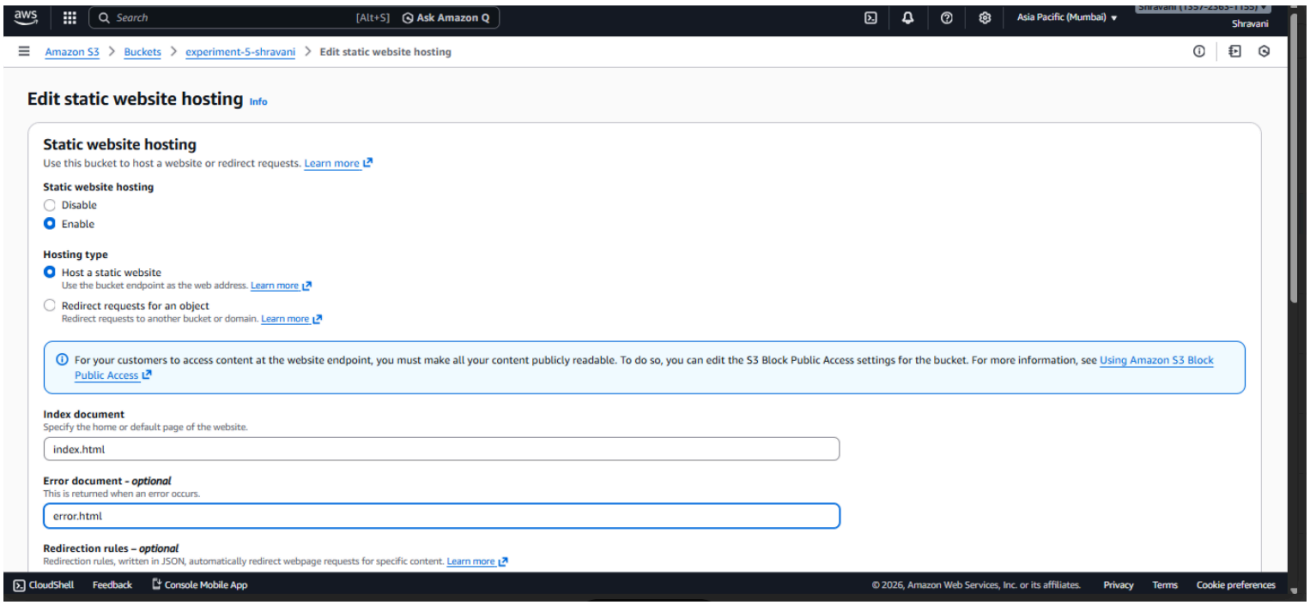
Step-17: Now for delete files click on checkbox of your file and then click on Delete Button. Write permanently delete and click on delete object button

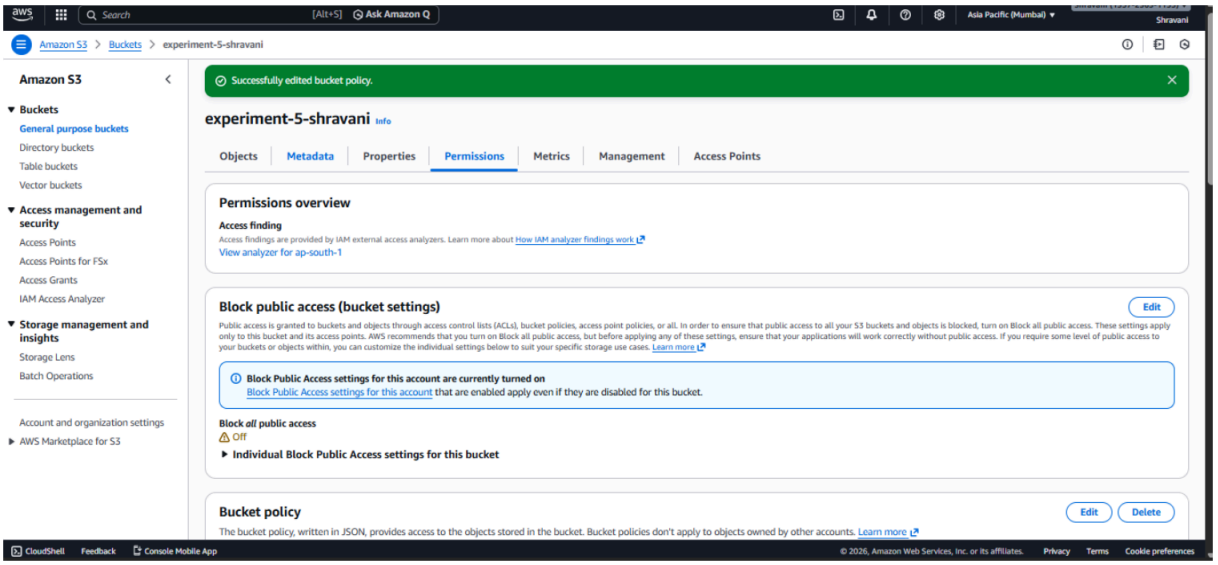
Now click on close button

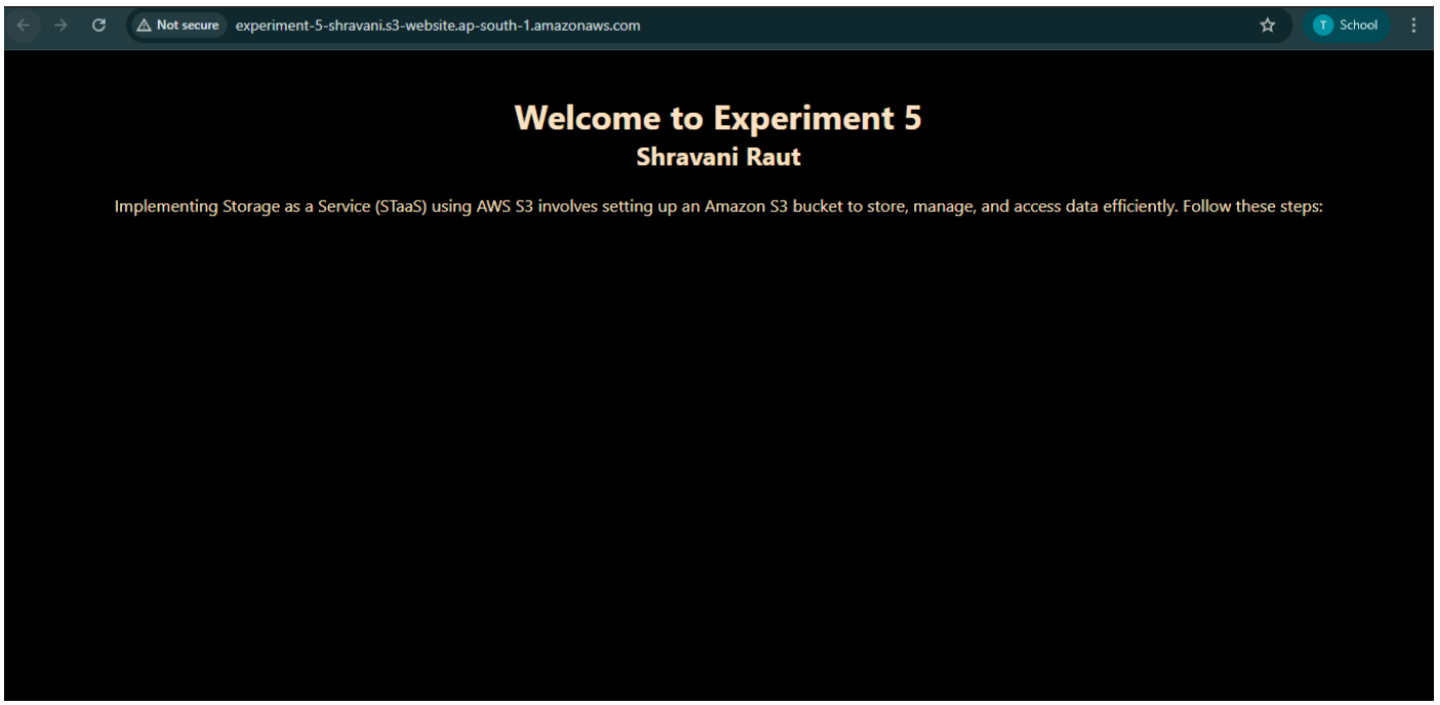
Step-18: now come to Amazon S3 tab and select your bucket and then click on delete button Write down your bucket name in delete bucket tab and click on delete button at bottom right You can see that the bucket is deleted.

Output/Observation:


Amazon S3 > Buckets > Create bucket
Create bucket Info
Buckets are containers for data stored in S3.
General configuration
AWS Region
Asia Pacific (Mumbai) ap-south-1
Bucket type Info
General purpose
Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.
Directory
Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.
Bucket name Info
experiment5-shravani
Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). Learn more
Copy settings from existing bucket - optional
Only the bucket settings in the following configuration are copied.
Choose bucket
Format: s3://bucket/prefix
Object Ownership Info
Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.
Object Ownership
ACLs disabled (recommended)
All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.
ACLs enabled
Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.


Amazon S3 > Buckets > experiment-5-shravani > Edit static website hosting
Edit static website hosting Info
Static website hosting
Use this bucket to host a website or redirect requests. Learn more
Static website hosting
Disable
Enable
Hosting type
Host a static website
Use the bucket endpoint as the web address. Learn more
Redirect requests for an object
Redirect requests to another bucket or domain. Learn more
For your customers to access content at the website endpoint, you must make all your content publicly readable. To do so, you can edit the S3 Block Public Access settings for the bucket. For more information, see Using Amazon S3 Block Public Access
Index document
Specify the home or default page of the website.
index.html
Error document - optional
This is returned when an error occurs.
error.html
Redirection rules - optional
Redirection rules, written in JSON, automatically redirect webpage requests for specific content. Learn more


Amazon S3 > Buckets > experiment-5-shravani
Successfully edited bucket policy.
experiment-5-shravani Info
Objects
Metadata
Properties
Permissions
Metrics
Management
Access Points
Permissions overview
Access finding
Access findings are provided by IAM external access analyzers. Learn more about How IAM analyzer findings work
View analyzer for ap-south-1
Block public access (bucket settings)
Edit
Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to all your S3 buckets and objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to your buckets or objects within, you can customize the individual settings below to suit your specific storage use cases. Learn more
Block Public Access settings for this account are currently turned on
Block Public Access settings for this account that are enabled apply even if they are disabled for this bucket.
Block all public access
Off
Individual Block Public Access settings for this bucket
Bucket policy
Edit
Delete
The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. Learn more



Conclusion:

Storage as a Service in Amazon Web Services (AWS) provides scalable, reliable, and cost-effective cloud storage solutions where users can store and access data over the internet without managing physical hardware. AWS offers services like Amazon S3 for object storage, Amazon EBS for virtual disk storage, and Amazon EFS for shared file systems. These services ensure high durability, security, automatic backup, and easy scalability, making them suitable for applications, backups, big data, and content storage while following a pay-as-you-use pricing model.